UMC 0.11um Logic and Mixed-Mode Al Advanced Enhancement 1.2V NCAP Device SPICE Model Document.

Version 0.5 Phase 1
2016/04/13

Table of Contents

List of Figures

List of Tables

1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.1_p1	2010/01/28	Kelly pao	(1) Original release based on new DSM definition form 2009/11/30.
0.1_p2	2010/03/04	Kelly pao	(1) Modified the description of nf range .
0.1_p3	2010/06/15	Kelly pao	(1) Modified the shrink methodology of Igate .
0.2_p1	2010/06/30	Kelly pao	(1) Based on the L110AE wafer. (2) Modified the corner ranges.
0.2_p2	2010/08/13	Kelly pao	(1) Modified Igate parameters.
0.3_p1	2010/10/19	Kelly pao	(1) Removed the description m in model cards. (2) Revised the Igate formula in model cards.
0.4_p1	2010/10/19	Kelly pao	(1) Revised the Igate formula in model cards
0.5_p1	2016/04/13	Kelly pao	(1) Revised the Igate parameter in spectre model cards

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

UMC 0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor Al Advanced Enhancement Process

1.3 Model Usage

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2014.9SP2
Cadence SPECTRE version 13.1.0.066
Mentor Graphics ELDO version 7.9_1.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below.

TYP: Typical NCAP case
MAX: Maximum NCAP case
MIN: Minimum NCAP case

Follow these steps to invoke the model file during simulation.

For Synopsys Hspice
- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.
.lib "./ModelFileName.lib" CaseName
, where CaseName could be TYP, MAX, or MIN.
- Define device condition by 'Xxx' statement.
Example:
X1 VGG GND DeviceName lf=2e-6 wf=5e-6 nf=4

For Cadence Spectre

- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.

include "./ModelFileName.lib.scs" section=CaseName

, where CaseName could be TYP, MAX, or MIN.

- Define device condition by 'Xxx' statement.

Example:

X1 (VGG GND) DeviceName lf=2e-6 wf=5e-6 nf=4

For Mentor Graphics Eldo

- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.

.lib "./ModelFileName.lib.eldo" CaseName

, where CaseName could be TYP, MAX, or MIN.

- Define device condition by 'Xxx' statement.

Example:

X1 VGG GND DeviceName lf=2e-6 wf=5e-6 nf=4

ModelFileName: l110ae_ncap12_v051

DeviceName: ncap_12

Geometry description (before 90% shrinkage):

0.15 μm <= Lf <= 2 μm

2.5 μm <= Wf <= 10 μm

Nf = 2~8

Bias Range:

-1.2 <= V_{GS} <= 1.2V(*1.1)

V_{DS} = V_{BS} = 0V

Temperature range:

-50 ℃ ~ +125 ℃

Layout example:

The following figure is the layout example of the NCAP. lf and wf are the single poly finger length and width, respectively, and nf is the total finger number.

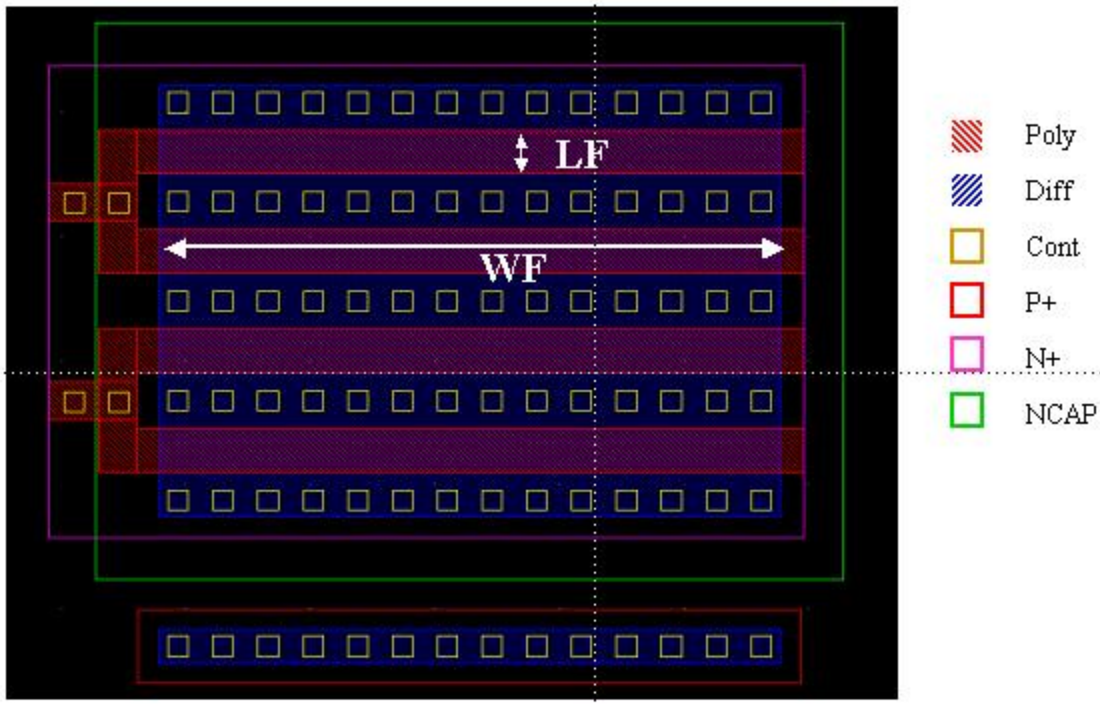


Figure 1-1 NCAP layout

2 Capacitance Model

2.1 Silicon Wafer for Modeling

The information of the hardware used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information for NCAP

	Wafer for NCAPs measurements data
Mask ID	BSW09A
Lot ID	D4QAC
Wafer Number	#5

2.1.1 Test devices

The following geometries were available for parameters extraction.

Table 2-2 Geometry of NCAP devices measured for parameters extraction.

wf\lf	0.147	0.462	0.912	1.812
9				x
4.5	x	x	x	x
2.25				x

2.1.2 Measurement conditions

The capacitance was measured under the conditions summarized in the table below.

Table 2-3 Test conditions for Cgg curves of NCAP

V_{DD}= 1.1V	C_G vs. V_G
V_G	From –1.32 to 1.32 V in 0.024 V step
V_B	0V
Temp [°C]	-50, 25, 125

2.2 Electrical Parameters

The following table shows the measured and simulated Cmax (fF/um^2), where Cmax is the capacitance at VG=1.1V, VB=0 V, T=25℃.

Table 2-4 Electrical Parameters for NCAP

All parameters are given for T=25 C, unless specified otherwise.

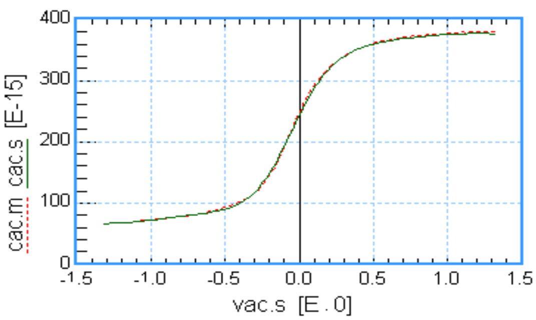
Name	LF (um)	WF (um)	Nf1	Model Cmax* (pF)
NCAP1_1	1.812	4.5	4	0.3774507
NCAP1_2	0.912	4.5	4	0.1915910
NCAP1_3	0.462	4.5	4	0.0978159
NCAP1_4	0.147	4.5	4	0.0293072
NCAP1_5	1.812	9	4	0.7511891
NCAP1_6	1.812	2.25	4	0.1921159
NCAP1_7	0.462	4.5	2	0.0482700
NCAP1_8	0.462	4.5	8	0.1930820

Note : wf & lf are after shrinkage sizes

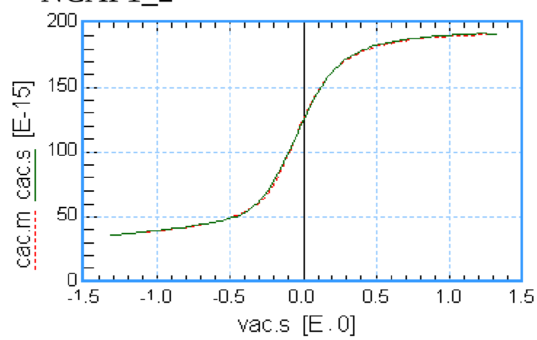
2.2.1 Cgg vs. Vg plots

The following figures show the measured and simulated Cgg (pF) vs. Vg (V). The measurement data is dots and simulation is line, respectively.

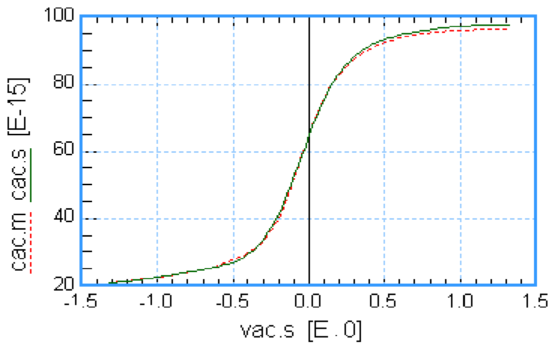
NCAP1_1



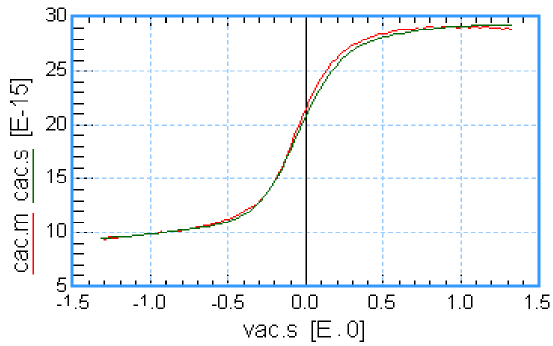
NCAP1_2



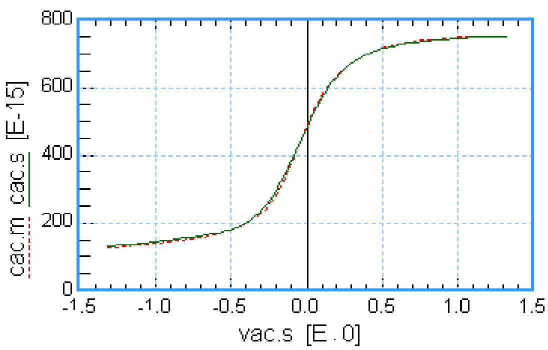
NCAP1_3



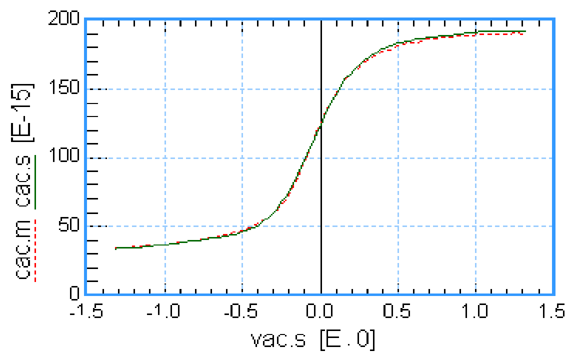
NCAP1_4



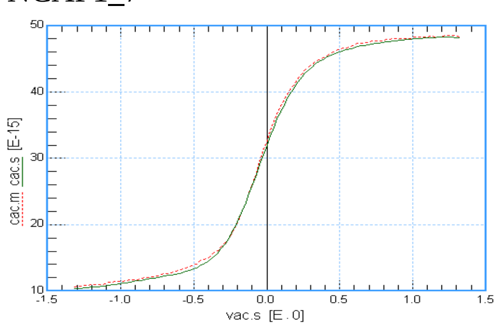
NCAP1_5



NCAP1_6



NCAP1_7



NCAP1_8

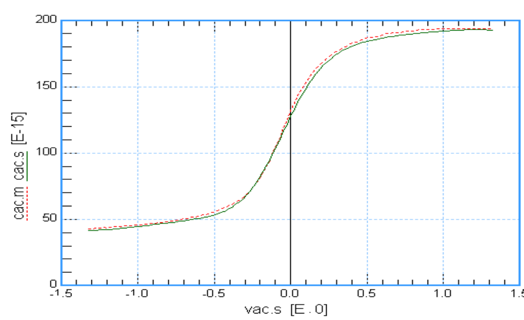


Figure 2-1 Cgg vs. Vg at 25 C

2.2.2 Ig vs. Vg plots

The following figures show the measured and simulated Ig (A) vs. Vg (V). The measurement data is dots and simulation is line, respectively.

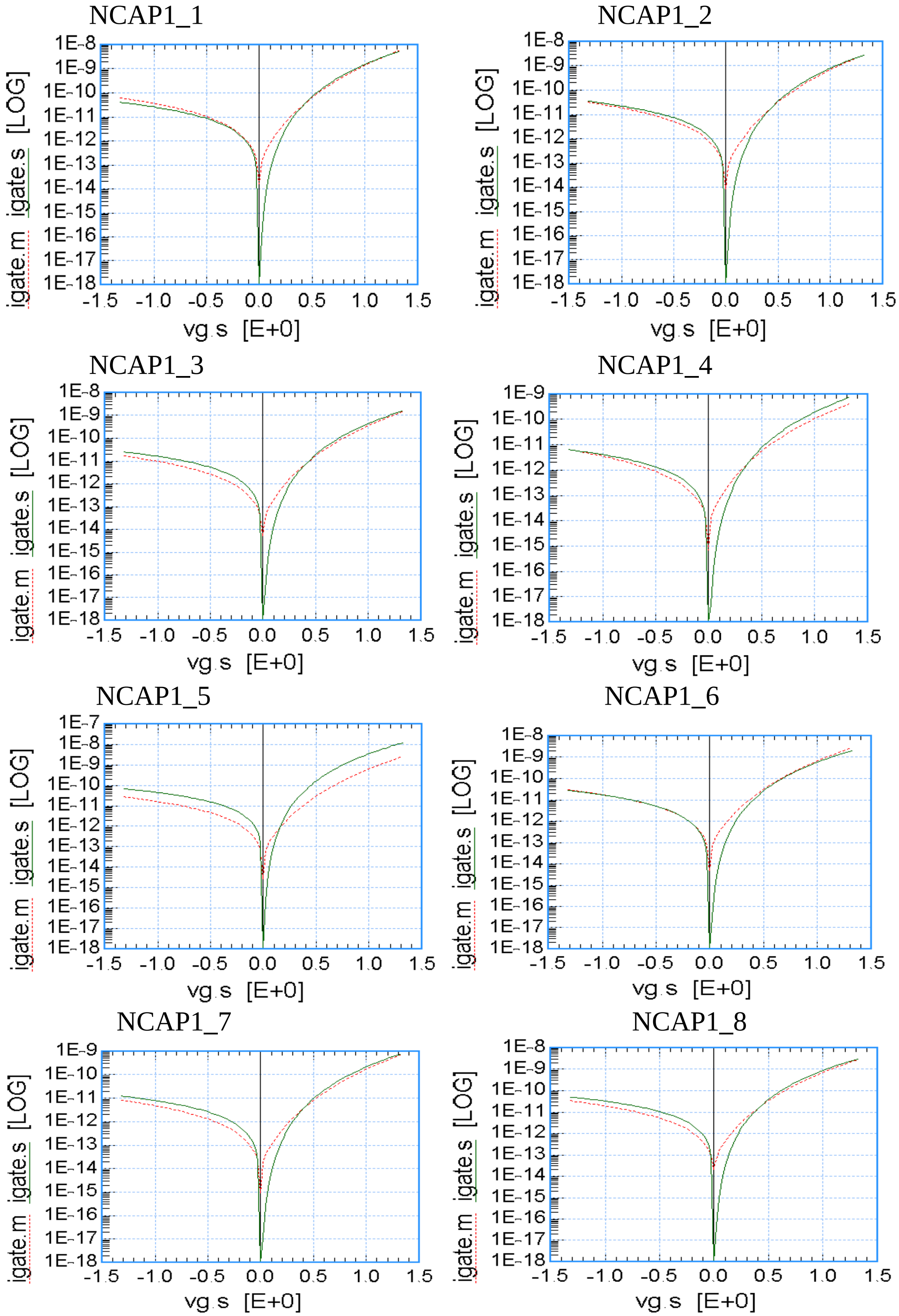


Figure 2-2 Ig vs. Vg at 25 C

3 Appendix

3.1 Hspice Warning Message

Table 3-1 Hspice warning message

NO.	Warning Message	Explanation
1.	None	None

3.2 ELDO Warning Message

Table 3-2 Eldo warning message

NO.	Warning Message	Explanation
1.	None	None

3.3 Spectre Warning Message

Table 3-3 Spectre warning message

NO.	Warning Message	Explanation
1	None	None